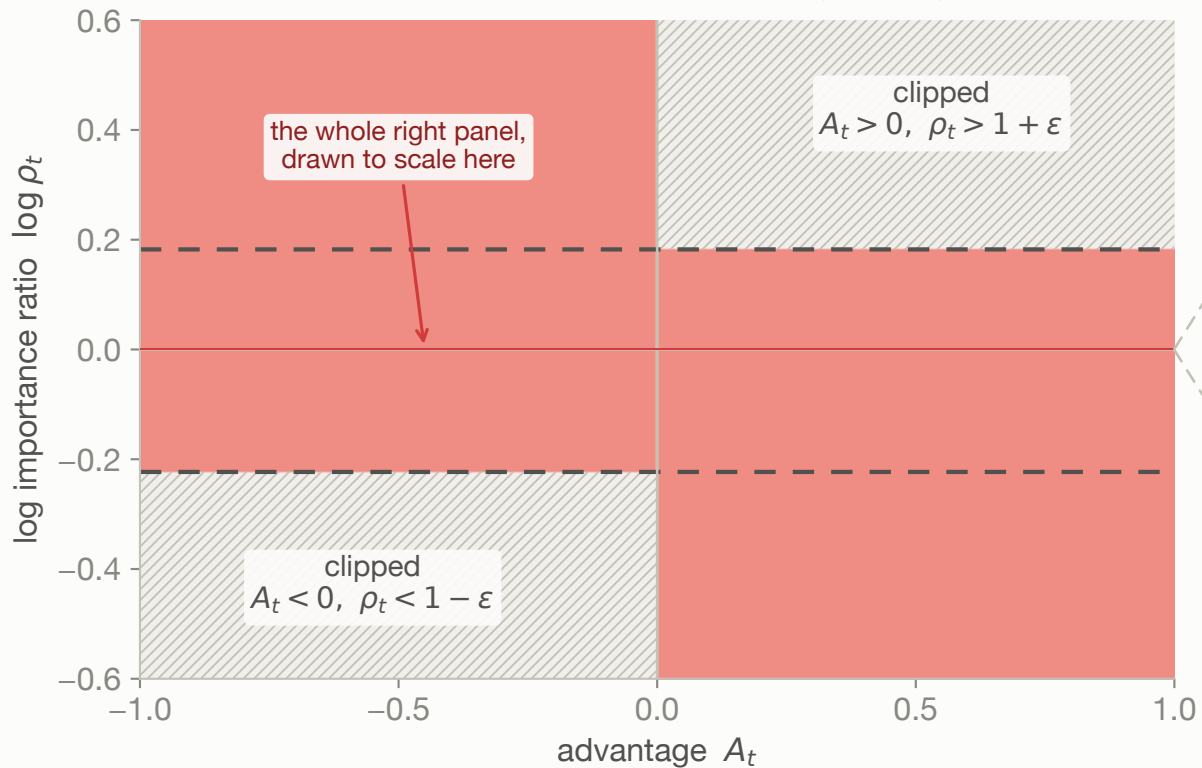


PPO / GRPO — token ratio ( $\varepsilon = 0.2$ )



GSPO — sequence ratio ( $\varepsilon_l = 3 \times 10^{-4}, \varepsilon_h = 4 \times 10^{-4}$ )

